Discussion of GLM exercise + mixed models

Example analysis

As an example, we will ask whether the local abundance of *Eulaema nigrita* depends on forest cover, while accounting for variation in yearly rainfall. We could write the analysis methods as follows:

To assess the response of *Eulaema nigrita* to variation in forest cover, we fitted a generalized linear model with negative binomial error distribution (accounting for severe overdispersion) and a log link function. To account for the expected greater abundance of the study species in drier areas, we included mean-centered mean annual precipitation as a covariate in the model.

```
dat$mcMAP = dat$MAP - mean(dat$MAP, na.rm=T)
m = glm.nb(Eulaema_nigrita ~ mcMAP + forest., data = dat)
summary(m)
```

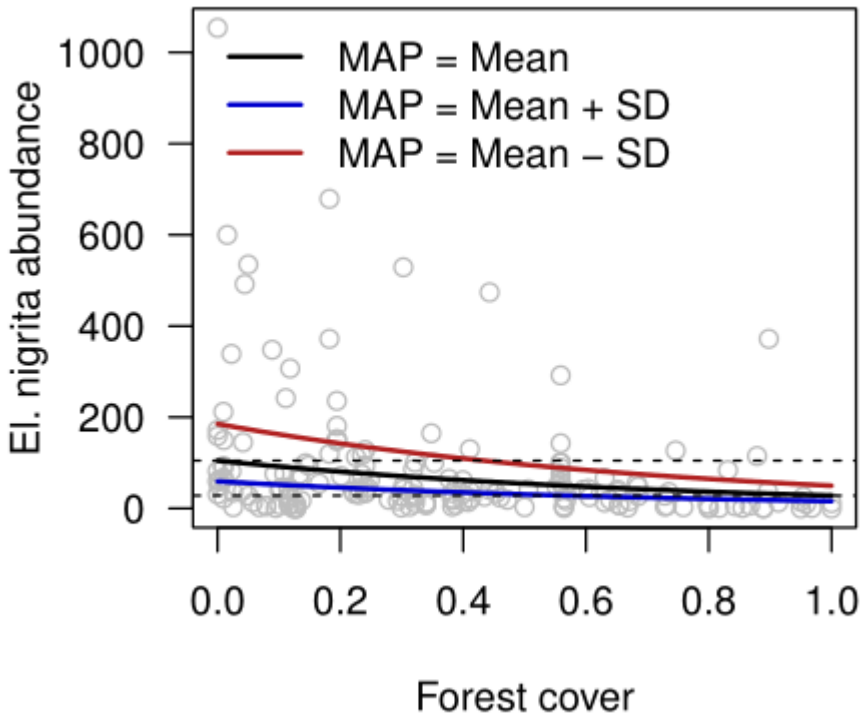
- To interpret the parameter estimates, it is best to backtransform the predicted values for some relevant scenarios to the original scale (here counts of bees).
- Here it made sense to mean-centre MAP but keep forest cover on the proportion scale, so that the intercept represents the scenario of “no forest but average rainfall”

```
coefs = summary(m)$coef
exp(coefs[1,1])
```

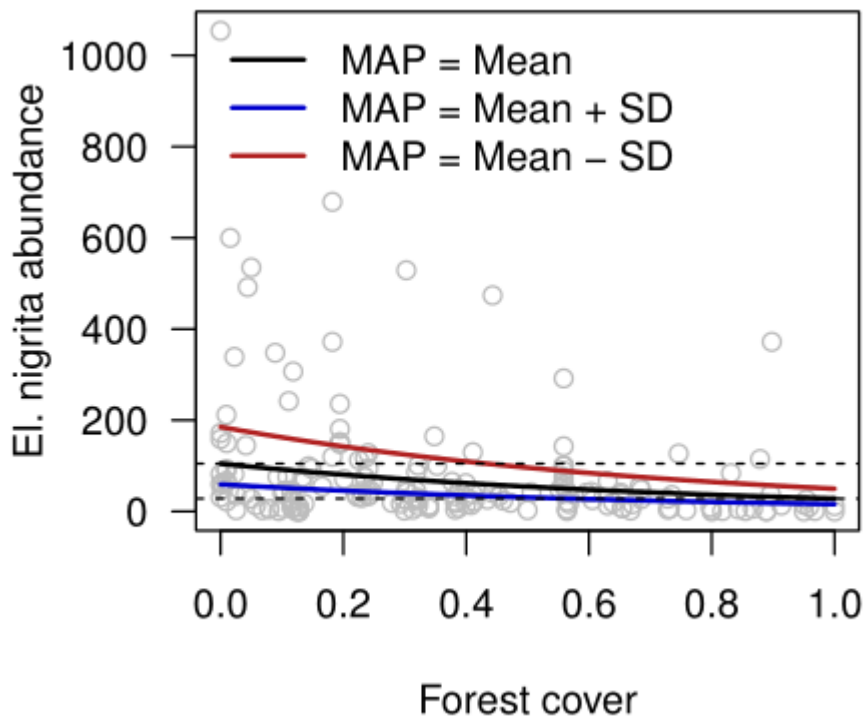
```
## [1] 104.958
```

```
exp(coefs[1,1] + coefs[3,1])
```

```
## [1] 28.26711
```



In the Results, we could then write "The abundance of *Eulaema nigrita* varied substantially along the forest-cover gradient (Fig. 1). Assuming average annual rainfall, predicted bee abundance ranged from 105 bees for a site in a completely unforested landscape to 28 bees in a completely forested landscape."



Typo in variance partitioning example

To interpret the actual variances, we could e.g. scale the standard deviations by the trait mean to obtain a coefficient of variation (CV). To maintain the additivity of the variances, an even better approach is to scale the variances themselves by the square of the trait mean, thus obtaining a squared CV, CV^2. We scale by the square of the mean because variances also have squared units, and we thus obtain a unitless number.

```
CV2_Among = VarAmongGroups/mean(y)^2
CV2_Within = VarWithinGroups/mean(y)^2
CV2_Total = CV2_Among + CV2_Within
```

Finally it is nice to organize all these values in a table.

```
df = data.frame(Mean = mean(y), SD = sd(y),
                Among = VarAmongGroups/(VarAmongGroups+VarWithinGroups)*100,
                Within = VarWithinGroups/(VarAmongGroups+VarWithinGroups)*100,
                CV2_Among, CV2_Within, CV2_Total)
df = apply(df, MARGIN=2, FUN=round, digits=3)
df
```

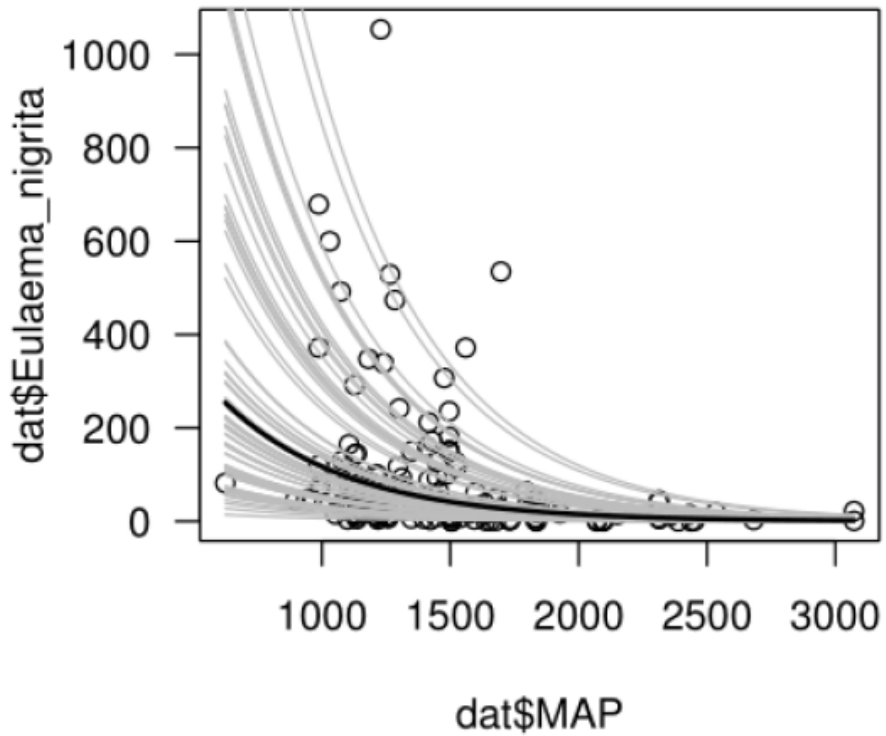
```
##       Mean         SD      Among     Within  CV2_Among CV2_Within  CV2_Total
##     38.768      4.602     43.680     56.320      0.006      0.008      0.014
```

Some clarifications on mixed models

- Random effects are always factors – not continuous
- With only two factor levels: usually fixed effect, especially if not a random sample of a much larger population (e.g. sex)

Generalized linear mixed models

```
##  Family: nbinom2  ( log )
## Formula:          Eulaema_nigrita ~ MAP + (1 | SA)
## Data: dat
##
##      AIC      BIC   logLik deviance df.resid
##   1790.8   1803.5   -891.4   1782.8      174
##
## Random effects:
##
## Conditional model:
##  Groups Name        Variance Std.Dev.
##  SA     (Intercept) 1.382    1.176
## Number of obs: 178, groups:  SA, 72
##
## Dispersion parameter for nbinom2 family (): 1.87
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  6.820199   0.542558  12.570  < 2e-16 ***
## MAP         -0.002057   0.000353  -5.827 5.65e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Processing and Analysis of Biological Data

BIOS15 2025

Lecture 8.
Model selection

$$y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i$$

Øystein H. Opedal

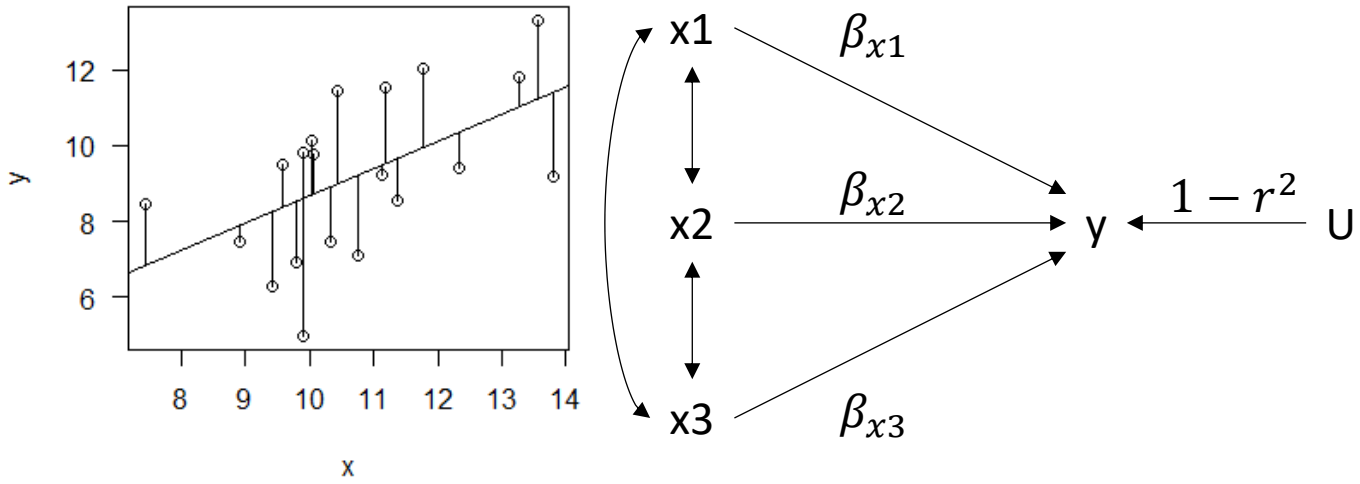
This week

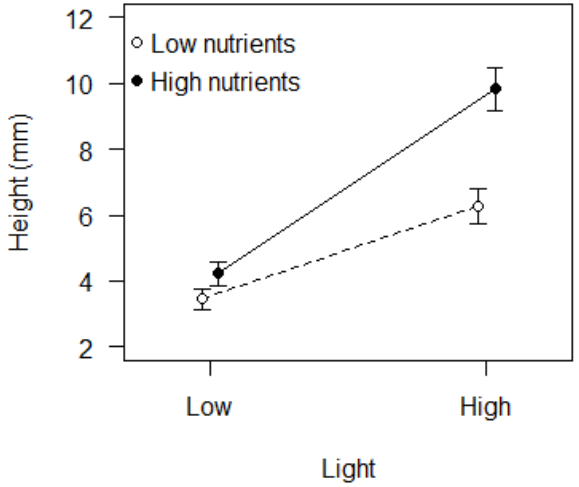
- Today
  - Lecture on **model selection**
  - Lecture on **generalized linear mixed models**
  - Seminar on **method validation**
  - Exercise session (model selection, GLMM)

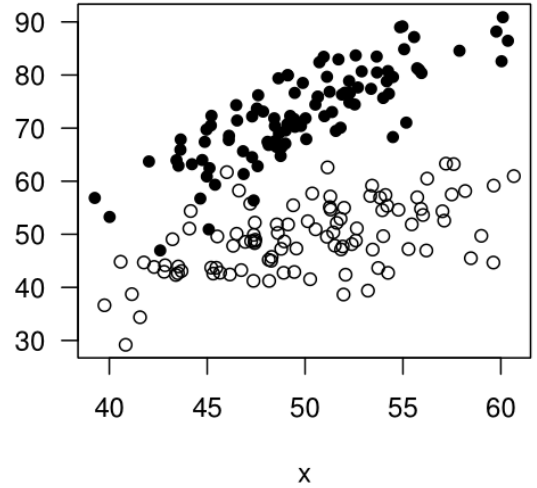
- Wednesday
  - Lecture on **multivariate methods**
  - Seminar
  - Exercise session (multivariate methods)

Overview of (generalized) linear (mixed) models

- Continuous covariates: (multiple) regression
- Categorical covariates: N-way ANOVA
- Continuous and categorical covariates: ANCOVA

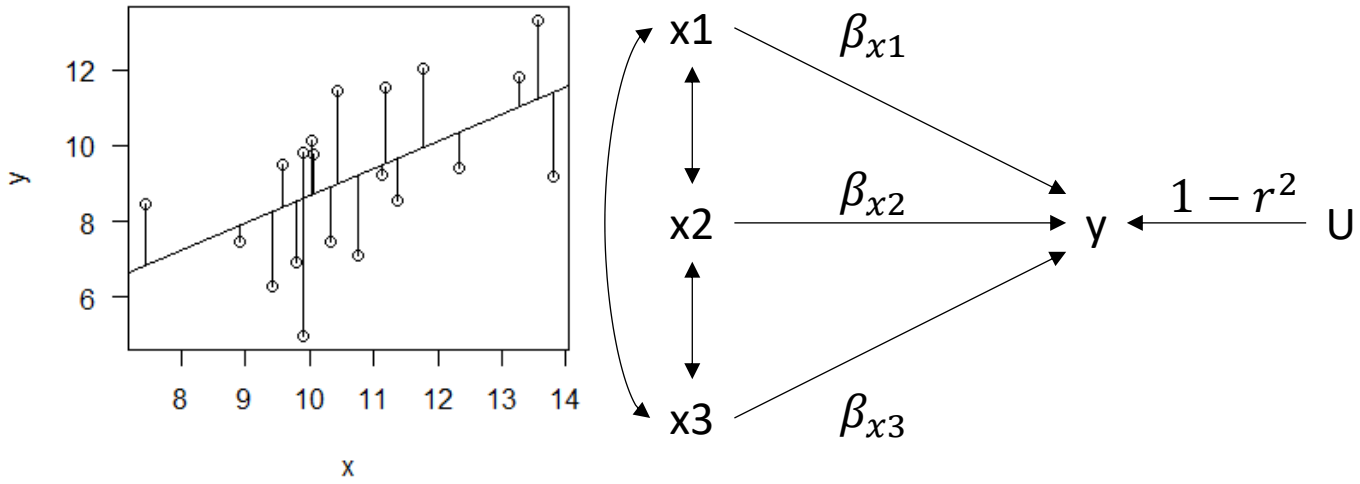


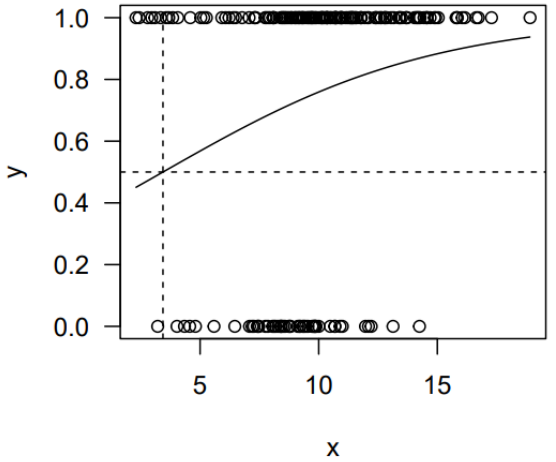


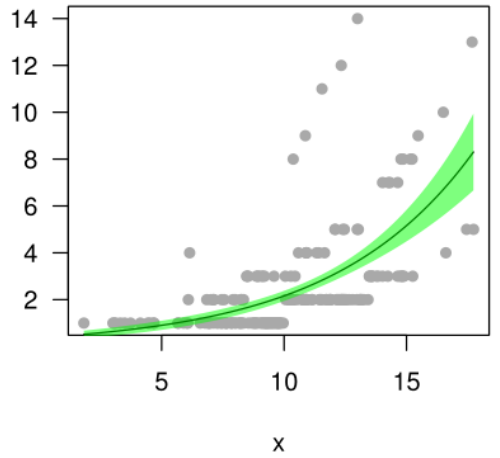


Overview of (generalized) linear (mixed) models

- Binary/proportional data: Logistic regression
- Count data: Poisson GLM
- Overdispersed count data: Negative binomial GLM



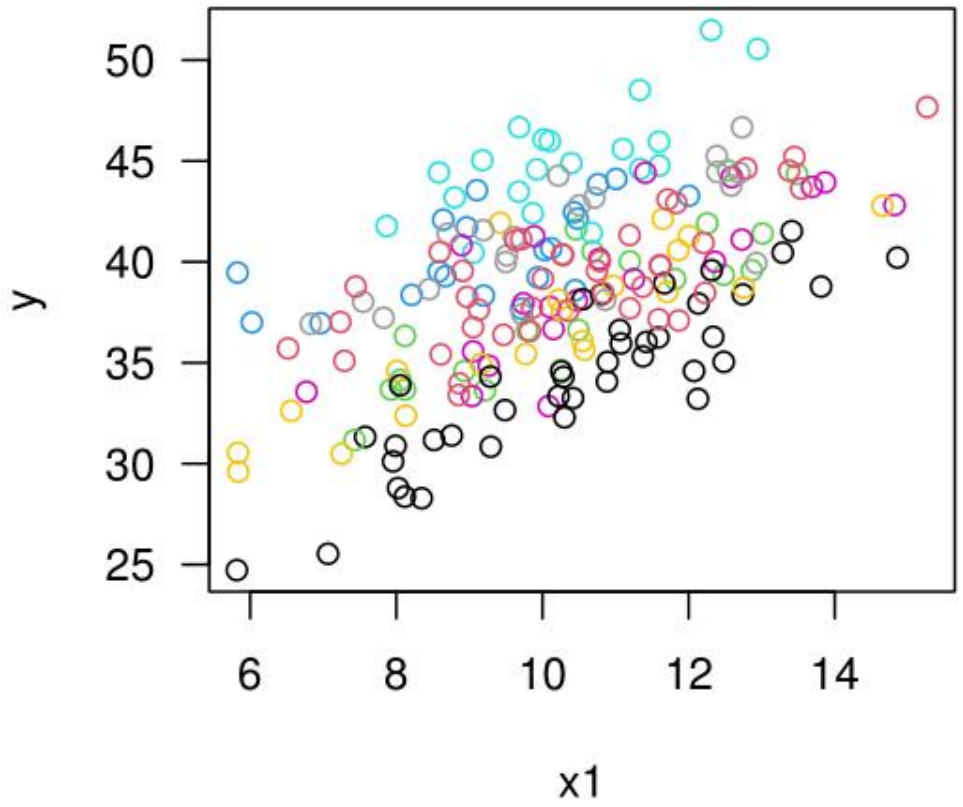




Fixed and random effects

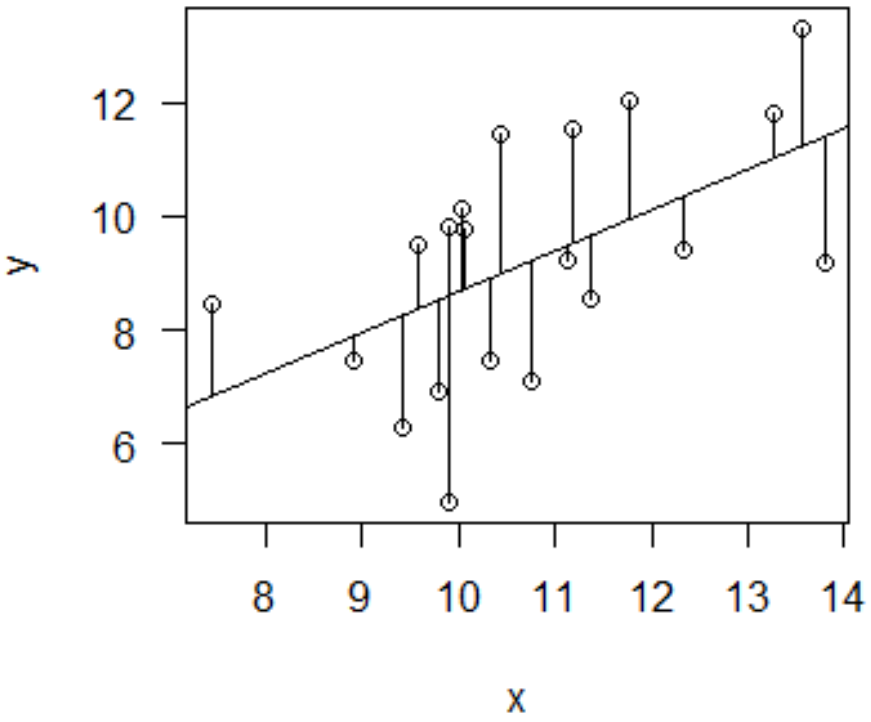
- Linear mixed models contain both fixed effects and **random effects**
- Fixed effects are those for which we are interested in the exact value of the response variable for each factor level
- Random effects are those for which we are more interested in the variance in the effect on the response variable.
- Levels of random effects are random simples from a larger population: rule of thumb is mínimum 6 levels
- Random effects are often used to account for hierarchical structure of sampling (e.g. plots nested in forest patches), but can also be used to estimate variance components

$$y = \mathbf{X}\beta + \mathbf{Z}\mu + \epsilon$$



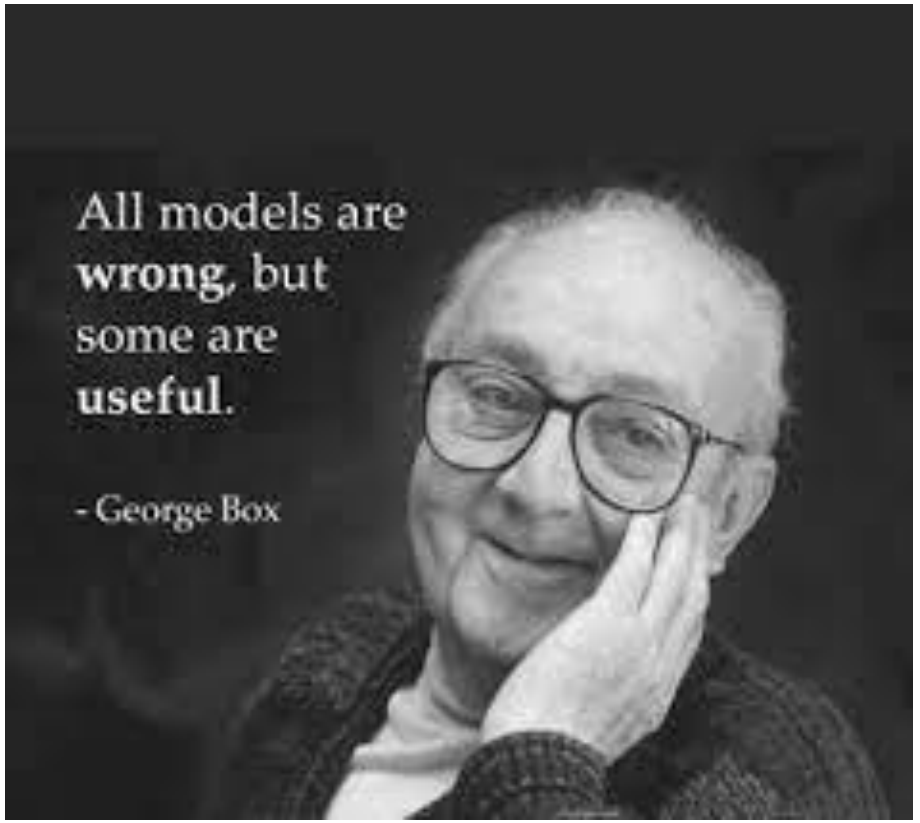
Measuring model fit

- One way to measure model fit is to ask how well the model explains variation in the response variable
- Linear models: r^2 (percent variation explained)
- Mixed models: r_M^2 (fixed effects only) and r_C^2 (fixed + random effects)
- Revisit how the r^2 is calculated in linear regression and ANOVA



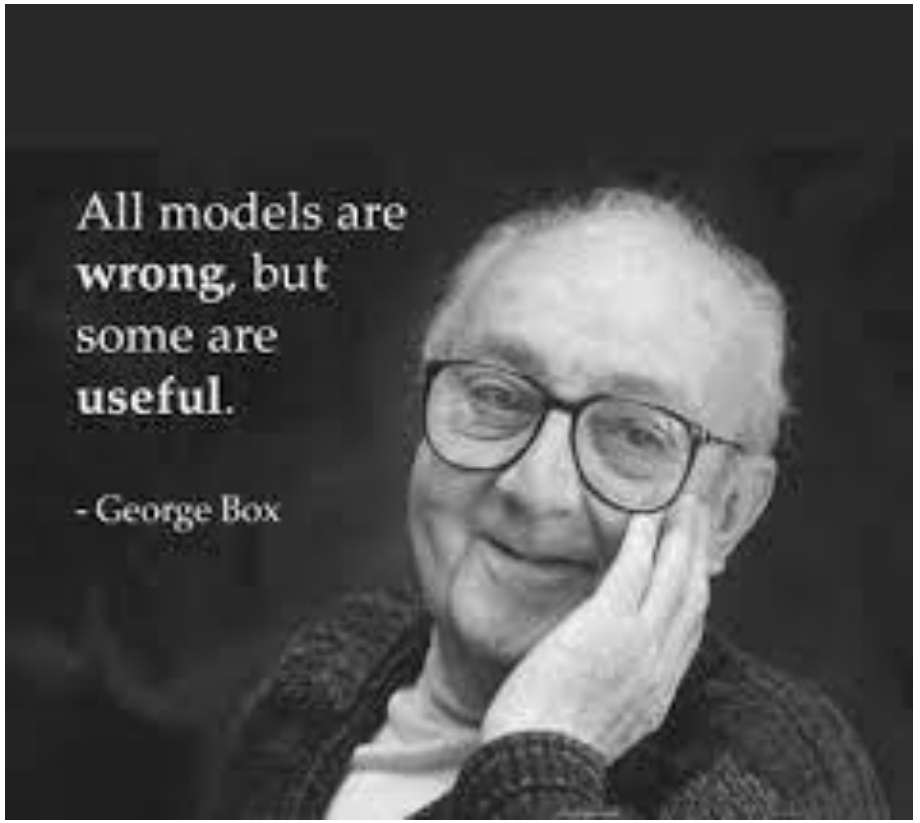
Confirmatory vs. exploratory analysis

- Often the parameters of interest are defined by the scope of the study. Then all we need to do is to fit the correct model and interpret the parameter estimates
- Sometimes we have a large set of candidate predictors and want to evaluate alternative models



Model selection

- Models could be chosen by sequentially removing non-supported terms until reaching a model with all terms supported (backwards selection, principle of parsimony, “Occam´s razor”)
- Problematic due to the issue with *p*-values, and due to disconnect from biological interpretation
- But sometimes we want to evaluate support for alternative models: model selection



Model selection with information criteria

- Information criteria are used to rank models in terms of their information content, roughly the goodness of fit achieved per parameter
- The Akaike Information criterion is defined simply as 2 times the log likelihood + a penalty of 2 per parameter.
- The highest ranked model is that with the lowest AIC value, and is thus chosen by trading off fit vs. simplicity
- We can compute the relative support for each model (Akaike weights).

$$AIC = -2ln(\hat{L}) + 2k,$$

$$w_i = \frac{exp(-\frac{1}{2}\Delta_i)}{\sum_{r=1}^{R} exp(-\frac{1}{2}\Delta_r)}$$

where the Δ values are the ΔAIC.

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where the Δ values are the ΔAIC.

```
##     df       AIC    logLik delta    w
## m2   4 1110.687 -551.3437  0.00 0.69
## m1   5 1112.549 -551.2745  1.86 0.27
## m3   3 1117.746 -555.8731  7.06 0.02
## m4   3 1119.242 -556.6211  8.55 0.01
## m5   2 1124.156 -560.0779 13.47 0.00
```

AIC model selection in mixed models

- We can select both random and fixed effects, but best to focus on one at a time
- To compare AIC values for models with different fixed effect, we need to fit the model by maximum likelihood (REML=F, the default in glmmTMB).
- Often recommended to refit the highest ranked model with restricted maximum likelihood (REML=T) to get parameter estimates

Writing (analysis) methods

- Normally a specific section at the end of the Methods (but sometimes integrated throughout)
- Focus on the aim of the analysis before technical details
- For models, list terms in words, sometimes include model equation
- R syntax is increasingly acceptable

Examples of model description

We analysed seed set per blossom by fitting linear mixed effect models with timing of pollination, pollen type and population, as well as their interactions, as fixed factors, and maternal individual as a random factor. Although seed number is a count variable, the distribution of residuals was close to normal, allowing for the use of a linear model. To analyse variation in seed mass, we fitted linear mixed effect models with timing of pollination, pollen type and population, as well as their interactions, as fixed factors. We further included seed set and peduncle diameter as covariates to account for a possible trade-off between seed mass and seed number within seed set, and possible blossom size effects, respectively. Blossom identity nested

To investigate the relationship between microclimate and plot richness, we fitted a mixed-effects Poisson regression model with species richness of the sample plots as the dependent variable, and microclimate variables as possible explanatory variables. To account for the structure of the data (sites nested within areas), site and area were entered as random factors. At the among-site scale (using each site

bruster et al. 2009b). We modeled pollen arrival during the female phase, $P_F(z')$, in units of pollen grains, as a function of bract area, gland area, and gland-stigma distance through a log link with a negative-binomial error distribution. The minimal model describing the predicted pollen load on the last day of the female phase was

$$P_F(z') = e^{a_2 + b_{21} UBA + b_{22} GA + b_{23} GSD}. \quad (2)$$

Writing results

- Start with summary statistics/patterns of variation

Field study

RESULTS

Phenotypic traits

In the field, average ASD varied nearly threefold among populations (range = 2.35–6.32 mm, Table S2), with 48.6% of the variance due to differences among populations (Table S4). Average gland size (17.43–34.62 mm^2) and average bract size (254.96–526.32 mm^2) varied approximately twofold among populations (Table S2), with differences among populations accounting for 28.1 and 19.9% of the variance respectively.

Meta-analysis/review

Database description

The updated database (Table S3) contains 792 evolvability estimates for 54 taxa representing 27 families. Among the 72 studies included, 68.6% were conducted on populations originating in North America (Fig. S1). Most studies were conducted in glasshouses or other controlled environments.

Writing results

- Start with summary statistics/patterns of variation
- Focus on biology over statistics (explain the result in biological terms)
- Quantify and exemplify (*y* increased by 0.5 mm per mm increase in *x*)

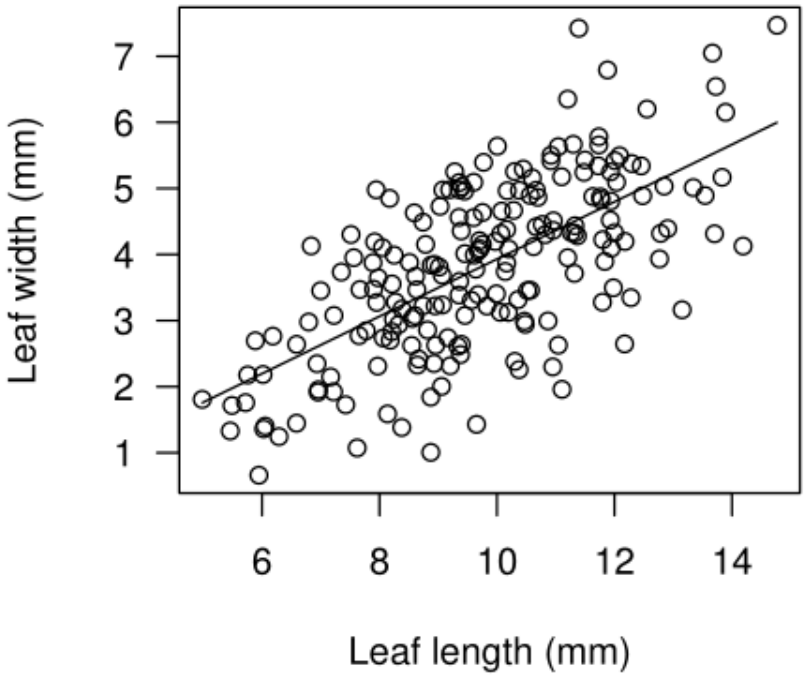
Overall, blossom development tended to be more rapid under dry conditions. In the wet treatment, the total receptive period of the blossoms (from the first day of opening to the abscision of the male cymule) lasted for 6.87 ± 0.09 d in the two large-glanded populations and 5.80 ± 0.10 d in the two small-glanded populations (fig. 4). In the dry treatment, the length of this period was moderately reduced by, on average, 5.4% across all four populations (fig. 4; table 4; see table A3 for model-selection results).

Writing results

- Start with summary statistics/patterns of variation
- Focus on biology over statistics (explain the result in biological terms)
- Quantify and exemplify (*y* increased by 0.5 mm per mm increase in *x*)
- No discussion (normally)
- Avoid unnecessary introductory sentences like “Population means are reported in Table 1”.
- Refer to tables and figures after a statement (“Body size varied substantially among populations (range = X.X to Y.Y, Table 1)”.

Making nice figures

- Labels and units!
- Keep regression lines within the data range
- If several symbols/colors: legend inside the graph
- Usually quite square panels look the best (not very rectangular)



There are also other packages for graphics

ggPlot



base R



Making nice tables

- No vertical lines
- All parameters explained in legend

Table 5

Inaccuracy Statistics under Wet and Dry Experimental Conditions

Taxon, population, treatment	Bias²	Male variance	Female variance	Joint inaccuracy (95% CI)	Mean-scaled inaccuracy (95% CI)
Large-glanded species:					
Tulum:					
Wet	.14 (16.1%)	.25 (28.8%)	.48 (55.1%)	.87 (.62, 1.17)	.03 (.02, .04)
Dry	.05 (3.0%)	1.00 (53.8%)	.80 (43.2%)	1.85 (1.42, 2.33)	.11 (.08, .14)
Puerto Morelos:					
Wet	.53 (25.8%)	.94 (46.0%)	.58 (28.2%)	2.05 (1.56, 2.66)	.07 (.05, .09)
Dry	1.42 (56.7%)	.69 (27.4%)	.40 (15.9%)	2.51 (1.94, 3.09)	.14 (.11, .17)
Small-glanded species:					
Cozumel:					
Wet	1.83 (63.7%)	.15 (5.1%)	.9 (31.3%)	2.87 (2.10, 3.71)	.22 (.17, .28)
Dry	.93 (50.2%)	.19 (10.5%)	.73 (39.4%)	1.85 (1.21, 2.57)	.20 (.14, .27)
Valladolid:					
Wet	.55 (38.8%)	.33 (23.5%)	.53 (37.7%)	1.41 (.82, 2.20)	.12 (.07, .19)
Dry	.29 (37.7%)	.24 (30.9%)	.24 (31.4%)	.77 (.43, 1.23)	.08 (.04, .13)

Note. Bias² is the mean squared deviation from the hypothesized adaptive optimum. The reported percentages are the proportion of the joint inaccuracy explained by each component. To obtain the mean-scaled inaccuracy, the joint inaccuracy was scaled by the product of the male and female trait means. Ninety-five percent confidence intervals (95% CIs) were obtained from 1000 nonparametric bootstrap estimates of the joint and mean-scaled inaccuracies, respectively.